

# Coupled TE–TM mode dynamics in a semiconductor laser subject to optical feedback: “the underdog’s triumph” scenario

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We investigate experimentally and theoretically the dynamics of polarization modes in a semiconductor laser subject to optical feedback exclusively into its dominant polarization mode. The laser, despite being edge emitting, weakly supports the orthogonally polarized mode even in a solitary configuration, i.e., without optical feedback. When subject to optical feedback, the laser displays transition behavior from correlated to anticorrelated dynamics between the polarization modes as a function of the injection current strength. We construct a theoretical model based on the Lang–Kobayashi rate equations to describe these results. We include coupling between the two polarization modes in the gain medium to reflect persistence of the second mode in both solitary configuration and under external feedback. With this coupling, the model can reproduce the observed experimental results. © 2015 Optical Society of America

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## 1. INTRODUCTION

A semiconductor laser (SCL) subject to delayed optical feedback is known to be a system that exhibits chaotic dynamics with a large variety of specific properties [1]. A particularly well-studied dynamical regime obtained for near-threshold driving currents and moderate optical feedback is known as the low-frequency fluctuations (LFF) regime. The LFF dynamics consists of two distinct time scales. The first, fast time dynamics, is characterized by short chaotic pulses of tens of picoseconds [2]. The second, significantly slower and still irregular power modulation dynamics, is responsible for the name LFF. It is characterized by sudden power dropouts followed by gradual stepwise revival to full power within a typical time of tens to hundreds of nanoseconds. Both time scales are well captured by a standard Lang–Kobayashi (LK) rate equations model [3]. Within this framework, Refs. [4,5] have proposed a theoretical interpretation of the LFF as a chaotic itinerancy with a drift between the local attractors associated with the cavity modes. The drift is toward the maximum gain mode (MGM) [6], in the vicinity of which the system spends most of the time. However, among the modes, MGM is the closest to a saddle-type antinode of the cavity, and an occasional collision (called crisis) between the local attractor and the antinode provokes a prolonged power dropout. After the crisis, the system restarts its itinerancy with a drift toward the MGM. Note that this schematic description of the LFF

regime leaves out a number of subtle theoretical issues (see Refs. [7,8]) that are beyond the scope of the present discussion. Here, we concentrate on studying the consequences of the dominant polarization mode power dropouts on the dynamics of the orthogonally polarized mode.

The dynamics between orthogonally polarized modes of edge-emitting SCLs under optical feedback have been intensively investigated in a particular type of setup where the cavity dominant TE mode is fed back to the orthogonally polarized, suppressed TM mode. Different chaotic regimes and correlation dynamics between the TE and TM modes have been reported in a number of experiments [9–13]. One such regime, which gives rise to robust square-wave polarization modulations, was recently discovered and characterized from both fundamental [14–17] and practical [18] points of view. Note, finally, that the fundamental dynamics of SCLs subject to optical feedback with variably rotated polarization has received little attention as yet and deserves further in-depth investigation [19,20].

The effect of polarization-rotated feedback has also been the subject of intense studies in vertical-cavity surface-emitting lasers, which are known to possess a polarization switching mechanism, attributed to the cylindrical symmetry of such devices [14,21–33].

The TM mode is usually strongly suppressed in edge-emitting SCLs. However, we detect its persistence in the whole

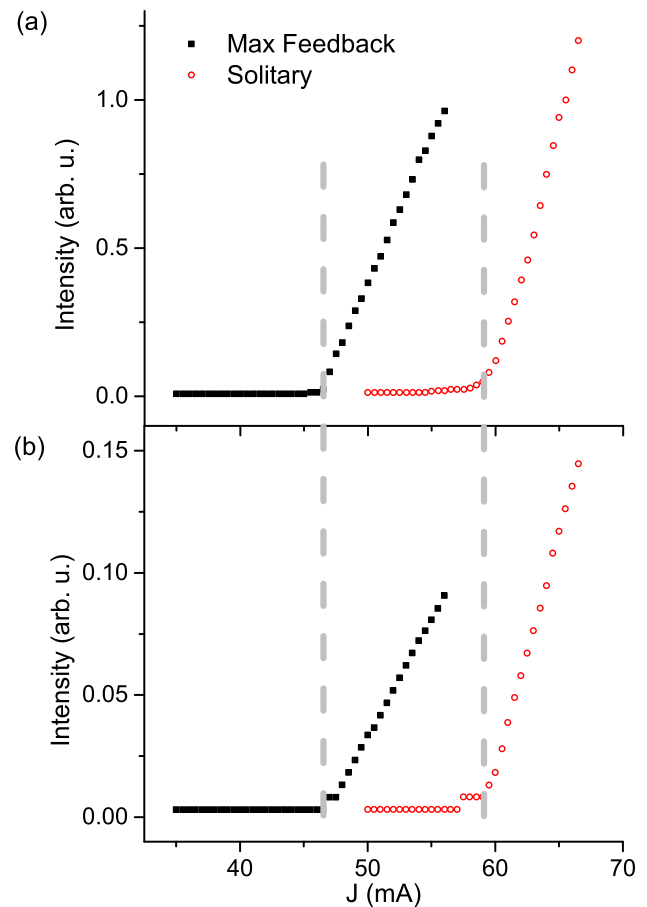
range of injection currents in a solitary configuration, i.e., when no external feedback is provided. Furthermore, in a setup where the dominant TE mode receives optical feedback while the weak TM mode does not, we show that the TM mode can be significantly enhanced as compared to its solitary lasing power during the phases of dominant mode collapse. This realizes “the underdog’s triumph” scenario in an SCL, in the sense that the weak mode of the laser becomes the dominant one for a short period of time. We stress that, in contrast to previous studies, the feedback’s polarization is not rotated but filtered and injected exclusively into the dominant mode of the laser. The observed behavior suggests the presence of coupling between the orthogonally polarized modes. Moreover, this coupling is independent from the strength of the external feedback and, thus, has to be introduced in the gain medium. To reproduce the polarization dynamics, we use the standard LK rate equations model, which we extend to include the intermode coupling. We verify that the coupling is necessary to support the persistent finite lasing power of the TM mode in both solitary configuration and under external feedback. Finally, we investigate the amount of coupling needed to reproduce the observed experimental results.

## 2. EXPERIMENTAL RESULTS

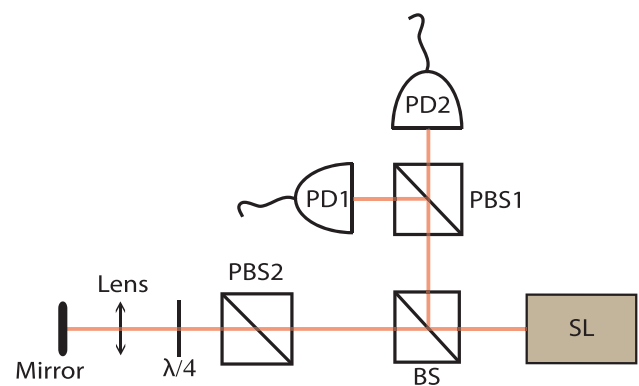
We use the Hitachi HL6545MGs diode edge-emitting SCL. The SCL is temperature stabilized to 25°C, where its central wavelength is  $\sim 661$  nm. The solitary laser’s threshold current is  $J_{th} = 58.6$  mA, and it is operated under a weak to moderate injection current regime characterized by  $p = J/J_{th} = 1.01 - 1.28$ . The nominal power of the laser is 120 mW and it supports both orthogonally polarized modes, although the power of the TM mode is significantly weaker, with a typical experimentally measured TE/TM power ratio of  $\sim 20$ . This observation hints at the presence of a lasing mechanism supporting the TM mode.

Moreover, when delayed optical feedback is selectively introduced to the dominant TE mode, the TM mode persists. In Fig. 1, the P-I curve of a laser under typical experimental conditions is shown. The presence of the distinctive threshold behavior in both modes demonstrates that the TM mode starts to lase at approximately the same current as the TE mode, but it always remains weak compared to the dominant mode. As both modes compete for the same pool of carriers and the TM mode is suppressed by the laser geometry, we attribute the TM mode’s lasing mechanism to coupling between the two modes inside the gain medium. This conclusion is also strongly supported by the character of the coupled mode dynamics that will be described next.

The experimental setup is shown schematically in Fig. 2. The laser beam is collimated with a short focal length lens positioned close to the output facet (not shown in Fig. 2). A regular beam splitter reflects  $\sim 8\%$  of the laser power for detection purposes. About 4% of the reflected power is used to detect each polarization mode independently by splitting the beam with a polarization beam splitter (PBS), denoted PBS1 in Fig. 2. Each mode’s time dynamics is then recorded with a fast photodiode (PD1 and PD2) on a Lecroy 64Xi 600 MHz bandwidth oscilloscope. An additional  $\sim 4\%$  of lasing power reflected from the



**Fig. 1.** P-I curves of the (a) TE and (b) TM modes of a typical laser used in the experiment. Red points (dark squares) describe the solitary (under external feedback) laser configuration and show reduced slope efficiency in both cases. Gray dashed vertical lines emphasize the existence of nearly the same threshold current for both TE and TM modes of the laser in both configurations.



**Fig. 2.** Schematic representation of the experimental setup. A polarization beam splitter (PBS1) allows for simultaneous detection of the TE and TM modes of the laser. PBS2 plays a double role in the setup. It prevents feedback into the TM mode and, simultaneously, in combination with a  $\lambda/4$  wave plate, controls the optical feedback strength reflected into the TE mode of the laser. A lens in the setup serves to improve mode-matching of the beam profile onto the external facet of the laser. BS is a regular beam splitter, reflecting  $\sim 8\%$  of the laser power for detection purposes; PD1 and PD2 are fast photodiodes.

second facet of the BS (not shown in Fig. 2) is used for the detection of the total lasing intensity with a slow detector. The main part of the beam passes through another PBS (denoted as PBS2) before being reflected back into the laser by a mirror positioned 2.5 m away from the laser, corresponding to a round-trip time of  $\tau = 16.7$  ns. PBS2 ensures exclusive TE mode delayed optical feedback. The optical feedback path includes a quarter-wave plate (a half-wave plate for the double-pass feedback light), which, in combination with PBS2, allows flexible tuning of the feedback power. A lens located at about a focal length distance in front of the mirror is mounted on an XYZ translation stage to improve mode-matching of the feedback beam profile onto the front facet of the SCL, considerably enhancing the feedback strength.

Our main experimental results are summarized in Fig. 3, which depicts the time dynamics of two laser modes for different values of  $p$ . The feedback strength is measured to correspond to 15% threshold reduction ( $1 - J_{\text{th}}/J_{\text{th}0}$ ) [34]. Our study is restricted to the regime of parameters where the dominant TE mode experiences LFFs, which can be readily identified in Fig. 3 (lower panels). However, the typical power collapses of the TE mode triggers quite a different dynamical

response of the weak TM mode, as seen in the upper panels of Fig. 3. For injection currents close to the solitary laser threshold current [Fig. 3(a)], the TM mode generally follows the power collapses of the TE mode. For larger  $p$  [Fig. 3(b)], power bursts start to develop in the TM mode at the edge of the TE mode collapses, and these anticorrelation dynamics of the two modes become more pronounced as the injection current  $p$  is increased [Fig. 3(c)]. This behavior persists for larger  $p$  until, eventually, the TM-mode LFFs almost disappear. The observed dynamics further supports the presence of coupling between the modes, and the external feedback cannot be responsible for it. We therefore conclude that the mode coupling is introduced in the gain medium. Based on this understanding, we build a theoretical model that is able to reproduce the observed behavior.

### 3. NUMERICAL MODEL

To analyze the TE–TM coupled mode dynamics we use the well-established LK rate equations model [3], which we extend to include two polarization modes [11], where only the dominant TE mode receives optical feedback. The rate equations describe the temporal evolution of the electrical fields  $E_1(t)$  (TE mode) and  $E_2(t)$  (TM mode), their respective phases  $\phi_1(t)$  and  $\phi_2(t)$ , and the carrier density above transparency  $n(t)$ :

$$\frac{dE_1}{dt} = \frac{1}{2}[n(t)G_1 - \gamma_1][E_1(t) + c_0E_2(t)] + \kappa E_1(t - \tau) \cos \Delta\phi, \quad (1)$$

$$\frac{d\phi_1}{dt} = \frac{\alpha}{2}[n(t)G_1 - \gamma_1] \left[ 1 + c_0 \frac{E_2(t)}{E_1(t)} \right] - \kappa \frac{E_1(t - \tau)}{E_1(t)} \sin \Delta\phi, \quad (2)$$

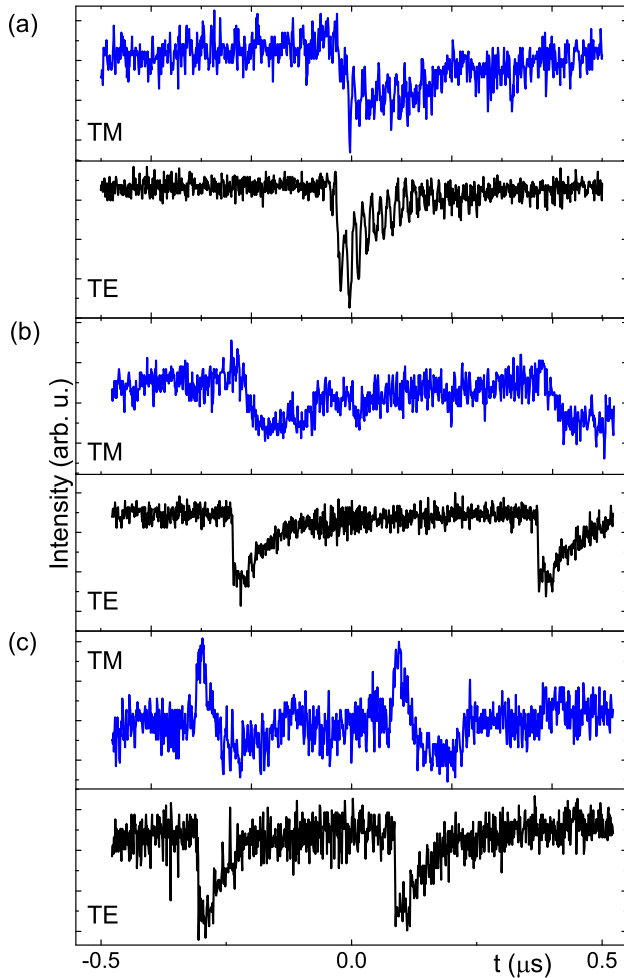
$$\frac{dE_2}{dt} = \frac{1}{2}[n(t)G_2 - \gamma_2][E_2(t) + c_0E_1(t)], \quad (3)$$

$$\frac{d\phi_2}{dt} = \frac{\alpha}{2}[n(t)G_2 - \gamma_2] \left[ 1 + c_0 \frac{E_1(t)}{E_2(t)} \right], \quad (4)$$

$$\frac{dn}{dt} = (p - 1)J_{\text{th}} - n(t)[G_1|E_1(t)|^2 + G_2|E_2(t)|^2 + \gamma_s], \quad (5)$$

where  $\Delta\phi = \omega\tau + \phi_{\text{TE}}(t) - \phi_{\text{TE}}(t - \tau)$ . Meanings and values of all parameters in Eqs. (1)–(5) are summarized in Table 1. The equations do not include the noise term due to spontaneous emission. We verified that such a term is incapable of reproducing experimentally observed behavior and does not affect the results of our model. We therefore exclude it from further consideration. Also, we assume the same frequency for both polarization modes. In general, they may be different, but the difference was measured to be below the resolution limit of our spectrum analyzer (<65 GHz). Introduction of small frequency difference in our simulations does not affect the results.

To reflect coupling between the polarization modes in the gain medium, we introduce a phenomenological coupling rate coefficient  $c_0$  in the model. When  $c_0 = 0$ , the equations reduce to their standard form [11] and the TM mode fails to persist for

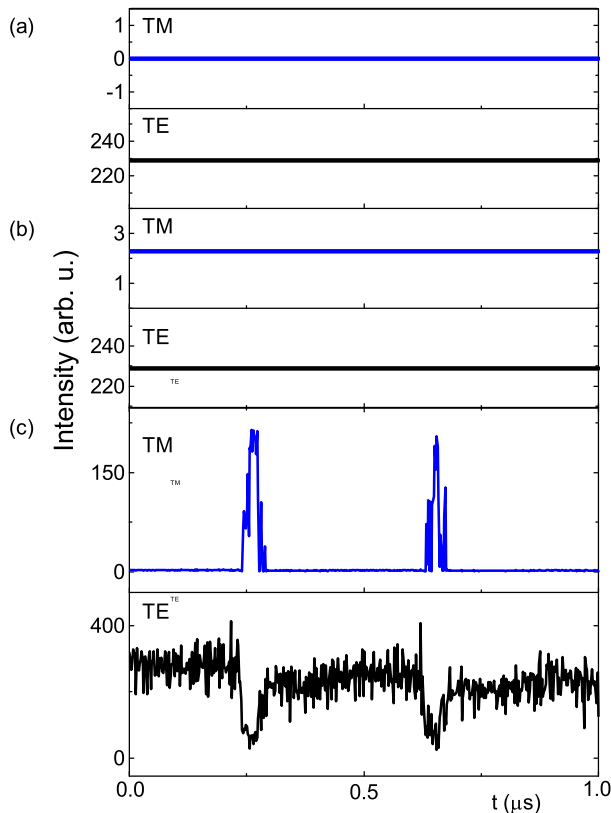


**Fig. 3.** Experimentally recorded time dynamics of the TM (TE) mode are represented in the upper (lower) panels for different  $p$  parameters: (a)  $p = 1.02$ , (b)  $p = 1.08$ , and (c)  $p = 1.13$ .

**Table 1. List of Parameters**

| Symbol          | Parameter                        | Value                                             |
|-----------------|----------------------------------|---------------------------------------------------|
| $G_1$           | TE mode gain coefficient         | $2.15 \times 10^3 \text{ m}^3 \text{ s}^{-1}$     |
| $G_2$           | TM mode gain coefficient         | $2.05 \times 10^3 \text{ m}^3 \text{ s}^{-1}$     |
| $\gamma_1$      | TE mode photon loss rate         | $3.6 \times 10^{11} \text{ s}^{-1}$               |
| $\gamma_2$      | TM mode photon loss rate         | $3.3 \times 10^{11} \text{ s}^{-1}$               |
| $\kappa$        | Feedback rate                    | $1 \times 10^{11} \text{ s}^{-1}$                 |
| $n_0$           | Carrier density at transparency  | $1.707 \times 10^8 \text{ m}^{-3}$                |
| $\gamma_s$      | Inverse of carrier lifetime      | $9.0 \times 10^8 \text{ s}^{-1}$                  |
| $\tau$          | External cavity roundtrip time   | $1.67 \times 10^{-8} \text{ s}$                   |
| $p$             | Injection current over threshold | 1–1.2                                             |
| $J_{\text{th}}$ | Injection current at threshold   | $1.247 \times 10^{18} \text{ m}^3 \text{ s}^{-1}$ |
| $\alpha$        | Linewidth enhancement factor     | 5.0                                               |
| $\lambda$       | Radiation wavelength             | 670 nm                                            |
| $c$             | Speed of light in vacuum         | $2.99 \times 10^8 \text{ ms}^{-1}$                |
| $\omega$        | Laser field frequency            | $2\pi c/\lambda$                                  |

any set of initial values. This is illustrated in Fig. 4(a), where we simulate this situation ( $c_0 = 0$ ) in the solitary laser configuration, i.e., with the external feedback rate set to zero ( $\kappa = 0$ ). Note that the TM mode remains essentially zero also for  $\kappa \neq 0$ . As both modes already share the same carrier reservoir [Eq. (5)], Fig. 4(a) clearly demonstrates that this direct coupling is not enough to explain our experimental results. However, the situation changes when coupling rate  $c_0$  is introduced. The



**Fig. 4.** TE–TM mode dynamics (a), (b) without and (c) with feedback, and (a) without and (b), (c) with ( $c_0 = 0.01$ ) coupling. For the TM mode to survive, the finite coupling has to be introduced as shown in (b) and (c). This condition is sufficient for the TM mode to persist even when the TE mode is subject to optical feedback, as in (c).

TM mode has finite steady-state power in the solitary laser configuration for  $c_0 = 0.01$ , as shown in Fig. 4(b). Figure 4(c) illustrates a typical dynamics for  $\kappa \neq 0$  demonstrating the persistence of the TM mode in this regime, as well. We, therefore, show that  $c_0 \neq 0$  is necessary to support the lasing of the TM mode in the solitary configuration and when the feedback is exclusively injected into the TE mode. Note that, if the TM mode receives the polarization rotated feedback from the dominant TE mode, as was shown in previous studies (such as Ref. [11]), the feedback itself acts as an effective coupling mechanism replacing the role of the coupling  $c_0$  parameter in our model. Let us stress also that we expect  $c_0$  to be device-dependent.

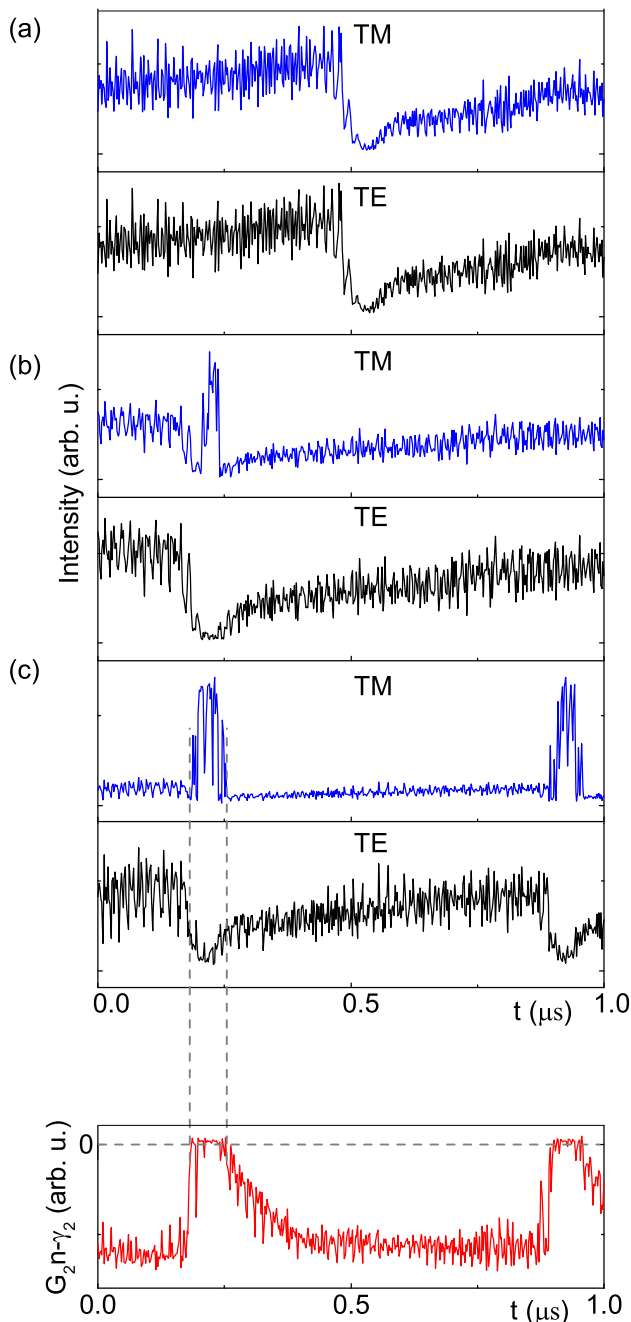
To reproduce the experimental results (see Fig. 3), we need to increase the value of  $c_0$ , as compared to the value used in Figs. 4(b) and 4(c). Generally, the coupling rate strength controls the value of the injection current ( $p$ ) for which the dynamics changes from correlated [Fig. 3(a)] to anticorrelated [Fig. 3(c)].

When  $c_0 \geq 0.5$ , there is no distinct anticorrelated dynamics, but a gradual transition from correlated dynamics to coherence collapse, and for weak enough coupling ( $c_0 \leq 0.01$ ), there is no apparent correlated dynamics, but only anticorrelated dynamics. We find that when  $c_0 \approx 0.1$ , the transition dynamics between the two regimes corresponds to experimental results presented in Fig. 3(b). This is shown in Fig. 5 (with  $c_0 = 0.1$ ), where the results were integrated over 2 ns to include the experimental time resolution due to the oscilloscope's bandwidth (600 MHz).

In the lower panels of Figs. 5(a)–5(c), the TE mode follows a typical LFF sequence for all  $p$  values. The corresponding dynamics of the TM mode is shown in the upper panels of Figs. 5(a)–5(c). For low  $p$  values ( $p \leq 0.02$ ), the TM mode correlates with the TE mode dynamics [Fig. 5(a)]. For larger values of the injection current  $p$ , power bursts in the TM mode become apparent each time the TE mode collapses [Fig. 5(b)]. For even larger  $p$  values, the power bursts of the TM mode increase further, and change the apparent mode dynamics from correlated to anticorrelated [Fig. 5(c)]. It is interesting to note that there are additional fine details in numerical simulations that agree with the experimental observations. For example, power bursts of the TM mode for moderate  $p$  are closely followed by local minima in TM power (residual LFF in the TM mode), which can be seen in experimental results, as well [Fig. 3(b)]. For larger  $p$ , this behavior nearly vanishes in both theory and experiment.

Duration of the TM-mode power bursts varies as a function of  $p$ . Fully developed bursts last a few tens of nanoseconds and are centered at the minima of the TE mode. In the LFF theory, when the TE mode collapses due to crisis, its recovery is delayed by a period of time equal to the round-trip time  $\tau$ . Apparently, the TM mode can burst for longer time periods and, in this respect, it is instructive to inspect the carrier density dynamics, which is shown in the lowest panel of Fig. 5 in the form of rate at transparency ( $G_2 n(t) - \gamma_2$ ) [see Eq. (3)]. This rate is negative most of the time, indicating that the carrier density is below that of the solitary laser. This situation is typical for SCLs subject to external optical feedback, which causes a decrease of the





**Fig. 5.** Numerically derived time dynamics of the TM (TE) mode are shown in the upper (lower) panels for different  $p$  parameters: (a)  $p = 1.02$ , (b)  $p = 1.08$ , and (c)  $p = 1.13$ . The lowest panel represents the carrier density dynamics in the form of the rate at transparency ( $G_2 n(t) - \gamma_2$ ) [see Eq. (3) and text]. Horizontal dashed line marks zero of this rate and vertical dashed lines limit the region where it becomes positive.

threshold current. However, for a short time, when the TE mode collapses, ( $G_2 n(t) - \gamma_2$ ) becomes positive and the gain medium is ready to support the normal operation of the laser. The usually dominant TE mode is suppressed due to the recent crisis. But the weak TM mode, which receives no direct feedback and “learns” about the crisis indirectly, through a sudden

change in the carrier density, can take over. For that short time, the laser’s usually strongly suppressed TM mode wins the mode competition and becomes the dominant mode of the laser.

#### 4. CONCLUSIONS

We investigated the behavior of an edge-emitting SCL that supports both TE and TM polarization modes and is subject to optical feedback exclusively in its dominant TE mode. We show that, in the regime of LFFs in the dominant TE mode, the TM mode demonstrates a nontrivial dynamics as a function of the strength of the injection current. When the injection current is increased, the TM-mode behavior switches from power drops in correlation with the TE mode, into anticorrelated power bursts. We present a theoretical model based on the LK rate equations. To reflect the fact that the polarization modes are coupled in the gain medium, we extend the standard model by adding a phenomenological coupling rate coefficient. The corrected model shows good agreement with the experimental results.

The interesting question for future research is to identify the exact physical mechanism behind the coupling term that we introduced here on a phenomenological basis. According to the model used in this paper, sharing the same carrier reservoir alone cannot explain the observed behavior. We suspect that such coupling can be caused by a crosstalk between the polarization modes within the laser’s internal cavity. As we experimentally show here, both modes lase in all configurations and create standing wave patterns within the cavity. However, counterpropagating orthogonally polarized beams support polarization gradients within the cavity, which can cause a mutual crosstalk between the modes.

The reported coupled-mode dynamics, in addition to being an interesting subject in its own right, can be useful in potential applications of chaotic SCLs. For example, weak current modulations in the vicinity of the reported transition can be used to encode information exclusively into the TM mode without affecting the dominant mode of the laser. Synchronized chaotic SCLs have been studied intensively in recent decades as communication devices for optical cryptography and as fast random number generators [1]. The TM mode can serve as an additional, multipurpose public communication channel in these schemes.

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#### REFERENCES

1. M. C. Soriano, J. Garcia-Ojalvo, C. R. Mirasso, and I. Fischer, “Complex photonics: dynamics and applications of delay-coupled semiconductor lasers,” *Rev. Mod. Phys.* **85**, 421–470 (2013).
2. I. Fischer, G. H. M. van Tartwijk, A. M. Levine, W. Elsässer, E. Göbe, and D. Lenstra, “Fast pulsing and chaotic itineracy with a drift in the coherent collapse of semiconductor lasers,” *Phys. Rev. Lett.* **76**, 220–223 (1996).
3. R. Lang and K. Kobayashi, “External optical feedback effects on semiconductor injection laser properties,” *IEEE J. Quantum Electron.* **16**, 347–355 (1980).
4. T. Sano, “Antimode dynamics and chaotic itineracy in the coherent collapse of semiconductor lasers with optical feedback,” *Phys. Rev. A* **50**, 2719–2726 (1994).

5. G. H. M. van Tartwijk, A. M. Levine, and D. Lenstra, "Sisyphus effect in semiconductor lasers with optical feedback," *IEEE Sel. Top. Quantum Electron.* **1**, 466–472 (1995).
6. A. Levine, G. Van Tartwijk, D. Lenstra, and T. Erneux, "Diode lasers with optical feedback: Stability of the maximum gain mode," *Phys. Rev. A* **52**, R3436 (1995).
7. A. Prasad, Y.-C. Lai, A. Gavrielides, and V. Kovanis, "Low-frequency fluctuations in external cavity semiconductor lasers: understanding based on a simple dynamical model," *J. Opt. B* **3**, 242–250 (2001).
8. D. Pieroux and P. Mandel, "Low-frequency fluctuations in the Lang-Kobayashi equations," *Phys. Rev. E* **68**, 036204 (2003).
9. J. Houlihan, G. Huyet, and J. G. McInerney, "Dynamics of a semiconductor laser with incoherent optical feedback," *Opt. Commun.* **199**, 175–179 (2001).
10. J. M. Saucedo-Solorio, D. W. Sukow, D. R. Hicks, and A. Gavrielides, "Bifurcations in a semiconductor laser subject to delayed incoherent feedback," *Opt. Commun.* **214**, 327–334 (2002).
11. T. Heil, A. Uchida, P. Davis, and T. Aida, "TE-TM dynamics in a semiconductor laser subject to polarization-rotated optical feedback," *Phys. Rev. A* **68**, 033811 (2003).
12. Y. Takeuchi, R. Shogenji, and J. Ohtsubo, "Chaotic dynamics in semiconductor lasers subjected to polarization rotated optical feedback," *Appl. Phys. Lett.* **93**, 181105 (2008).
13. Y. Takeuchi, R. Shogenji, and J. Ohtsubo, "Chaos dynamics in semiconductor lasers with polarization-rotated feedback," *Opt. Rev.* **17**, 144–151 (2010).
14. A. Gavrielides, T. Erneux, D. Sukow, G. Burner, T. McLachlan, J. Miller, and J. Amonette, "Square-wave self-modulation in diode lasers with polarization-rotated optical feedback," *Opt. Lett.* **31**, 2006–2008 (2006).
15. A. Gavrielides, D. W. Sukow, G. Burner, T. MacLachlan, J. Miller, and J. Amonette, "Simple and complex square waves in an edge-emitting diode laser with polarization-rotated optical feedback," *Phys. Rev. E* **81**, 056209 (2010).
16. G. Friart, L. Weicker, J. Danckaert, and T. Erneux, "Relaxation and square-wave oscillations in a semiconductor laser with polarization rotated optical feedback," *Opt. Express* **22**, 6905–6918 (2014).
17. G. Friart, G. Verschaffelt, J. Danckaert, and T. Erneux, "All-optical controlled switching between time-periodic square waves in diode lasers with delayed feedback," *Opt. Lett.* **39**, 6098–6101 (2014).
18. N. Oliver, M. C. Soriano, D. W. Sukow, and I. Fischer, "Dynamics of a semiconductor laser with polarization-rotated feedback and its utilization for random bit generation," *Opt. Lett.* **36**, 4632–4634 (2011).
19. S. Ramanujan, G. P. Agrawal, J. M. Chalwek, and H. Winful, "Elliptical polarization emission from GaAlAs laser diodes in an external cavity configuration," *IEEE J. Quantum Electron.* **32**, 213–221 (1996).
20. L. Khaykovich, T. Galfsky, Z. Shotan, and N. Gross, "TE-TM coupled mode dynamics in a semiconductor laser subject to feedback with variably rotated polarization," *Opt. Commun.* **282**, 2059–2061 (2009).
21. M. San Miguel, Q. Feng, and J. Moloney, "Light-polarization dynamics in surface-emitting semiconductor lasers," *Phys. Rev. A* **52**, 1728–1739 (1995).
22. J. Martin-Regalado, F. Prati, M. San Miguel, and N. Abraham, "Polarization properties of vertical-cavity surface-emitting lasers," *IEEE J. Quantum Electron.* **33**, 765–783 (1997).
23. G. Ropars, P. Langot, M. Brunel, M. Vallet, F. Bretenaker, A. L. Floch, and K. D. Choquette, "Experimental evidence of single round-trip oscillation in polarization self-modulated vertical-cavity surface emitting lasers," *Appl. Phys. Lett.* **70**, 2661–2663 (1997).
24. F. Robert, P. Besnard, M. L. Charles, and G. M. Stephan, "Polarization modulation dynamics of vertical-cavity surface-emitting lasers with an extended cavity," *IEEE J. Quantum Electron.* **33**, 2231–2239 (1997).
25. H. Li, A. Hohl, A. Gavrielides, H. Hou, and K. Choquette, "Stable polarization self-modulation in vertical-cavity surface-emitting lasers," *Appl. Phys. Lett.* **72**, 2355–2357 (1998).
26. M. Giudici, S. Balle, T. Ackemann, S. Barland, and J. R. Tredicce, "Polarization dynamics in vertical-cavity surface-emitting lasers with optical feedback: experiment and model," *J. Opt. Soc. Am. B* **16**, 2114–2123 (1999).
27. M. Sciamanna, T. Erneux, F. Rogister, O. Deparis, P. Megret, and M. Blondel, "Bifurcation bridges between external-cavity modes lead to polarization self-modulation in vertical-cavity surface-emitting lasers," *Phys. Rev. A* **65**, 041801 (2002).
28. M. Sciamanna, C. Masoller, F. Rogister, P. Mégret, N. Abraham, and M. Blondel, "Fast pulsing dynamics of a vertical-cavity surface-emitting laser operating in the low-frequency fluctuation regime," *Phys. Rev. A* **68**, 015805 (2003).
29. J. Mulet, M. Giudici, J. Javaloyes, and S. Balle, "Square-wave switching by crossed-polarization gain modulation in vertical-cavity semiconductor lasers," *Phys. Rev. A* **76**, 043801 (2007).
30. S. Y. Xiang, W. Pan, L. S. Yan, B. Luo, N. Jiang, and L. Yang, "Polarization properties of vertical-cavity surface-emitting lasers subject to feedback with variably rotated polarization angle," *Appl. Opt.* **48**, 5176–5183 (2009).
31. S. Y. Xiang, W. Pan, L. S. Yan, B. Luo, X. H. Zou, N. Jiang, and K. H. Wen, "Variable-polarization optical feedback induced hysteresis of the polarization switching in vertical-cavity surface-emitting lasers," *J. Opt. Soc. Am. B* **27**, 2512–2517 (2010).
32. S. Y. Xiang, W. Pan, L. S. Yan, B. Luo, X. H. Zou, N. Jiang, K. H. Wen, X. H. Zou, and L. Yang, "Polarization degree of vertical-cavity surface-emitting lasers subject to optical feedback with controllable polarization," *J. Opt. Soc. Am. B* **27**, 476–483 (2010).
33. H. Aoyama, S. Tomida, R. Shogenji, and J. Ohtsubo, "Chaos dynamics in vertical-cavity surface-emitting semiconductor lasers with polarization-selected optical feedback," *Opt. Commun.* **284**, 1405–1411 (2011).
34. N. Gross, W. Kinzel, I. Kanter, M. Rosenbluh, and L. Khaykovich, "Synchronization of mutually versus unidirectionally coupled chaotic semiconductor lasers," *Opt. Commun.* **267**, 464–468 (2006).